

HIV Antiviral Agents - Factoid Sheet

Week 8 • Core Lecture 8 • Anderson deck controls

Life-cycle map: entry → reverse transcription → integration → assembly/maturation. Match the class to the step before memorizing the drug.

ART Goals + Starting Considerations

Goals

Undetectable viral load • no replication = less mutation • U=U • CD4 recovery • fewer opportunistic infections • longer life expectancy • less transmission.

Before choosing ART

ART history/naivety • comorbidities • baseline viral load • drug-drug interactions • HIV resistance • adherence/convenience • pregnancy/breastfeeding.

Classes at a Glance

Class	Target / mechanism	High-yield toxicity / interaction
NRTI	Reverse transcription; handout: phosphorylated faulty building blocks → DNA chain termination	Older mitochondrial toxicity; drug-specific TDF kidney/bone, abacavir HLA-B*5701
NNRTI	Reverse transcriptase; handout: allosteric enzyme inhibition	Rash/SJS, hepatitis, low resistance barrier, CYP interactions
INSTI	Blocks insertion of HIV DNA into host genome	Generally well tolerated; weight gain/insomnia/depression; polyvalent-cation chelation
PI	Blocks protease cleavage → immature/nonfunctional virions	Hyperglycemia, GI intolerance, lipodystrophy; CYP3A4 + boosters
Entry	CCR5 / CD4 / gp120 / fusion	Agent-specific; often reserved for resistant virus
Lenacapavir	Capsid-dependent nuclear transport + viral/capsid assembly	Injection-site reaction, nausea, headache

NRTI Must-Knows

Drug	Pearl
TDF	Fanconi / nephrotoxicity + ↓ bone mineral density
TAF	Same toxicity pattern, less severe in deck
Abacavir	HLA-B*5701 hypersensitivity • never rechallenge • cardiovascular events • no HBV activity
FTC / 3TC	Well tolerated backbone partners; M184V causes resistance to both
Zidovudine	Older toxicities; perinatal prophylaxis. Handout: macrocytic anemia / marrow suppression.

NNRTI / INSTI / PI Recognition

Drug/class	Recognition clue
Rilpivirine	Meal + acid-suppressor interaction; depression/headache/rash/QTc prolongation
Efavirenz	Vivid dreams, dizziness, insomnia; empty stomach on deck
INSTIs	High barrier; best tolerated/efficacious class on deck; cation chelation
Bictegravir	May raise creatinine via tubular secretion without glomerular-function change
Elvitegravir	Take with food; requires booster
Cabotegravir	Long-acting injectable; injection-site pain
PIs	Ritonavir/cobicistat boosting; metabolic syndrome + GI intolerance; CYP interactions

Entry, Regimens, Start & Prevention

Entry / Capsid Agents

Agent	Mechanism / pearl
Maraviroc	CCR5 antagonist; requires tropism test ; only works for CCR5-tropic virus
Ibalizumab	Monoclonal antibody binds CD4 D2 region; post-attachment inhibitor; IV
Fostemsavir	gp120 attachment inhibitor; nausea/LFT elevation/sleep disturbance/QTc prolongation
Lenacapavir	Capsid inhibitor; nuclear transport + viral/capsid assembly; oral or IM; treatment/prevention
Enfuvirtide	<i>Handout enrichment:</i> gp41 fusion inhibitor; SQ injection-site reactions

What / When to Start

Deck: start ART as soon as possible regardless of CD4 → rapid entry • Test & Treat • treatment as prevention.

Current-deck framework	Note
INSTI + 2 NRTIs	Listed option
INSTI + 1 NRTI	Also listed - overrides older handout simplification
Boosted PI + 2 NRTIs	Listed option
NNRTI + 2 NRTIs	Listed option
Pregnancy	Deck names bictegravir/dolutegravir as preferred INSTIs; darunavir/ritonavir as preferred PI
HBV coinfection	Generally include 2 NRTIs with anti-HBV activity; flare possible if stopped/switched

Rapid-start exceptions on deck: cryptococcal meningitis • tuberculosis meningitis.

IRIS

Unmasking

Previously occult infection becomes apparent after immune recovery.

Paradoxical

Known/treated infection appears to worsen after immune recovery.

Prevention

Concept	Use
U=U / TasP	Suppressed viral load prevents sexual transmission
PrEP	Before exposure; deck says daily or intermittent; efficacy rises with adherence
PEP	After exposure; occupational exposure / sexual assault etc.

Handout practical PEP: 3-drug ART for 28 days; start ASAP and no later than 72 h after significant exposure.

Resistance + Exam Traps

Clue	Answer
NNRTI resistance	K103N
Lamivudine / emtricitabine resistance	M184V
PI / INSTI resistance	Multiple mutations generally required
HLA-B*5701	Avoid abacavir
Fanconi + bone loss	TDF
Vivid dreams	Efavirenz
Polyvalent cations	INSTI chelation
Cushing-like lipodystrophy + hyperglycemia	Protease inhibitor
Tropism assay first	Maraviroc
Needlestick after HIV exposure	PEP, not PrEP

Last 30 seconds: entry = maraviroc/ibalizumab/fostemsavir • RT = NRTI/NNRTI • integration = INSTI • maturation = PI • capsid = lenacapavir.

Primary: attached Anderson Antiretroviral Agents deck. Secondary: attached older-numbered HIV Antiretrovirals handout. Current resource title/position: Week 8 Core Lecture 8, HIV Antiviral Agents.